

Santhosh Phanitalpak Gandhala

Pune, Maharashtra, India | gsanthoshphanitalpak@gmail.com | +91 8179439677

Portfolio: <https://gsphanitalpak.github.io/gspanitalpak-portfolio> | **GitHub:** <https://github.com/gspanitalpak>

PROFESSIONAL SUMMARY

Embedded AI & Computer Vision Engineer with hands-on experience in machine learning, real-time image processing, sensor integration, and Raspberry Pi/Arduino-based intelligent systems. Built computer vision, IoT, and automation prototypes using Python, OpenCV, TensorFlow, and PyTorch. IEEE/Springer published researcher with strong interest in robotics, edge AI, and hardware-software integrated systems.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, JavaScript, SQL, Bash
- **AI / Machine Learning:** TensorFlow, PyTorch, scikit-learn, OpenCV, CNNs, NLP, Transfer Learning, BERT/RoBERTa, LSTM, Statistical ML
- **Embedded Systems:** Arduino, Raspberry Pi, Embedded C, GPIO Programming, Sensor Integration, Serial Communication, IoT Systems, Real-Time Sensor Pipelines
- **Robotics & Simulation:** ROS2 (Learning), Gazebo, SLAM concepts, Embedded Linux, Sensor Fusion basics, Navigation stack fundamentals
- **Computer Vision & Edge AI:** Image Processing, Facial Recognition, Real-Time Inference, Video Processing, Object Detection, Sensor Fusion, Edge AI Prototyping
- **Cloud / DevOps / Tools:** AWS, Docker, Linux, GitHub Actions, Raspbian, Fritzting, TinkerCAD

EXPERIENCE

Robotics & STEM Instructor — Entecres Labs Pvt. Ltd. (Ryan International Academy) | Jun 2025 - Nov 2025

- Designed and guided robotics, embedded systems, and sensor-based engineering projects using Arduino, Raspberry Pi, motors, actuators, and real-time data workflows.
- Mentored students in building automation prototypes involving sensors, embedded logic, and hardware-software integration.
- Conducted hands-on prototyping sessions focused on engineering thinking, debugging, and practical system development.

Full-stack Developer Intern & SME ICT — Byline Learning Solutions LLP | Dec 2024 – Apr 2025 | Pune

- Developed executable robotics and embedded system projects using Arduino, Raspberry Pi, Python, and sensor integration workflows.
- Built and tested classroom-deployable IoT and automation prototypes.
- Reviewed and debugged Python-based robotics and embedded codebases.
- Contributed to robotics-focused technical content and practical implementation modules.

Full stack Developer Intern| R&D Intern — M/s Technospirit | Jan 2024 – May 2024 | Hyderabad

- Built Raspberry Pi-based embedded prototypes involving GPIO interfacing and automation logic.
- Developed Flask-based interfaces for hardware-software communication and real-time data interaction.
- Worked on embedded simulation workflows, circuit prototyping, and system debugging.
- Improved frontend system usability for internal engineering dashboards.

FEATURED PROJECTS

StressSense — Edge AI Stress Monitoring System - TensorFlow, PyTorch, Raspberry Pi, Python, SwiftUI

- Designed and Built an IoT-enabled physiological real-time sensor acquisition and inference for monitoring and stress prediction system integrating GSR, MAX30102, and temperature sensors with Raspberry Pi.
- Developed ML pipelines for stress prediction using TensorFlow, PyTorch, and ensemble models.

Facial Recognition System - OpenCV, Python, Flask, SVM

- Developed a real-time facial recognition pipeline involving camera capture, preprocessing, face embeddings, and identity classification.
- Implemented confidence filtering, live inference workflows, and Flask-based operator interface with repeated testing under different real-world circumstances.

Driver Drowsiness Detection System - Arduino, IR Sensors, GSM Module

- Developed an embedded alert system for detecting driver fatigue using sensor-triggered logic and real-time alert mechanisms.
- Integrated actuator response and GSM-based notification workflows.

Vision-Guided Robotic Sorting Prototype - Python, OpenCV, Raspberry Pi

- Developed a camera-assisted object detection and robotic sorting workflow integrating visual input with actuator-based response logic.
- Focused on practical constraints such as false detection reduction, object differentiation, and synchronization between perception and action systems.

EDUCATION: Bachelor of Technology — Computer Science & Engineering (Hons. Cloud Computing)

Symbiosis Institute of Technology, Pune | 2020 – 2024

PUBLICATIONS

7 IEEE/Springer indexed publications in AI, Robotics, IoT, Advanced ML. ([indexed research papers link](#))

LANGUAGES: English (Business) | Hindi (Business) | Telugu (Native) | Japanese (N5)